Probability-Addition and Multiplication Rule

Objectives

- Discuss how addition and multiplication in probability works.
- Identify the rules in probability addition and multiplication.
- Solve problems pertaining to addition and multiplication probability.

Addition Rules for Probability

Experiment: A single 6-sided die is rolled. What is the probability of rolling a 2 or a 5?

Possibilities:

1. The number rolled can be a 2.
2. The number rolled can be a 5.

Events: *These events are mutually exclusive since they cannot occur at the same time.*

Probabilities: How do we find the probabilities of these mutually exclusive events? We need a rule to guide us.

Addition Rules for Probability

- **Addition Rule 1:** *When two events, A and B, are mutually exclusive, the probability that A or B will occur is the sum of the probability of each event.*
- P(A or B) = P(A) + P(B)

Let's use this addition rule to find the probability for Experiment 1.

- Experiment 1: A single 6-sided die is rolled. What is the probability of rolling a 2 or a 5?

Addition Rules for Probability

Probabilities:

$$P(2) = \frac{1}{6}$$

$$P(5) = \frac{1}{6}$$

$$P(2 \text{ or } 5) = P(2) + P(5)$$

$$= \frac{1}{6} + \frac{1}{6}$$

$$= \frac{2}{6}$$

$$= 1$$

Addition Rules for Probability

- Experiment 2: A spinner has 4 equal sectors colored yellow, blue, green, and red. What is the probability of landing on red or blue after spinning this spinner?

Probabilities:

$$P(\text{red}) = \frac{1}{4}$$

$$P(\text{blue}) = \frac{1}{4}$$

$$\begin{aligned} P(\text{red or blue}) &= P(\text{red}) + P(\text{blue}) \\ &= \frac{1}{4} + \frac{1}{4} \\ &= \frac{2}{4} \\ &= \frac{1}{2} \end{aligned}$$

Addition Rules for Probability

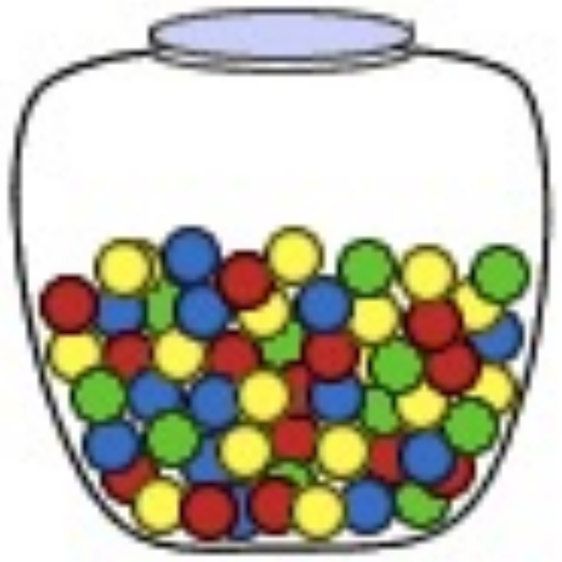
- Experiment 3: A glass jar contains 1 red, 3 green, 2 blue, and 4 yellow marbles. If a single marble is chosen at random from the jar, what is the probability that it is yellow or green?

Probabilities:

$$P(\text{yellow}) = \frac{4}{10}$$

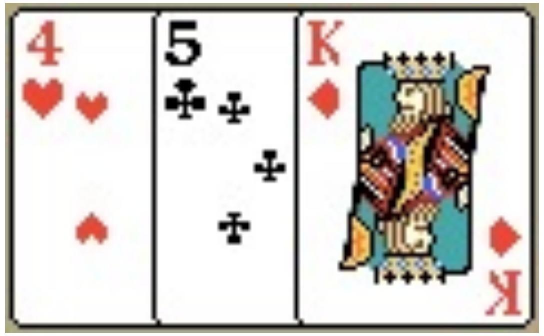
$$P(\text{green}) = \frac{3}{10}$$

$$\begin{aligned} P(\text{yellow or green}) &= P(\text{yellow}) + P(\text{green}) \\ &= \frac{4}{10} + \frac{3}{10} \\ &= \frac{7}{10} \end{aligned}$$



- In each of the three experiments above, the events are mutually exclusive. Let's look at some experiments in which the events are non-mutually exclusive.

- Experiment 4: A single card is chosen at random from a standard deck of 52 playing cards. What is the probability of choosing a king or a club?



Probabilities:

$$
\begin{aligned}
P(\text{king or club}) &= P(\text{king}) + P(\text{club}) - P(\text{king of clubs}) \\
&= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} \\
&= \frac{16}{52} \\
&= \frac{4}{13}
\end{aligned}
$$

- In Experiment 4, the events are non-mutually exclusive. The addition causes the king of clubs to be counted twice, so its probability must be subtracted. When two events are non-mutually exclusive, a different addition rule must be used.

Addition Rules for Probability

Additional Rule 2: When two events, A and B, are non-mutually exclusive, the probability that A or B will occur is:

- P(A or B) = P(A) + P(B) - P(A and B)

In the rule above, P(A and B) refers to the overlap of the two events.

Addition Rules for Probability

- Experiment 5: In a math class of 30 students, 17 are boys and 13 are girls. On a unit test, 4 boys and 5 girls made an A grade. If a student is chosen at random from the class, what is the probability of choosing a girl or an A student?

Probabilities: P(girl or A) = P(girl) + P(A) - P(girl and A)

$$= \frac{13}{30} + \frac{9}{30} - \frac{5}{30}$$

$$= \frac{17}{30}$$

Addition Rules for Probability

- Experiment 6: On New Year's Eve, the probability of a person having a car accident is 0.09. The probability of a person driving while intoxicated is 0.32 and probability of a person having a car accident while intoxicated is 0.15. What is the probability of a person driving while intoxicated or having a car accident?

Probabilities:

P(intoxicated or accident)	=	P(intoxicated)	+	P(accident)	-	P(intoxicated and accident)
	=	0.32	+	0.09	-	0.15
	=	0.26				

Summary

To find the probability of event A or B, we must first determine whether the events are mutually exclusive or non-mutually exclusive. Then we can apply the appropriate Addition Rule:

- Addition Rule 1: When two events, A and B, are mutually exclusive, the probability that A or B will occur is the sum of the probability of each event.

$$P(A \text{ or } B) = P(A) + P(B)$$

- Addition Rule 2: When two events, A and B, are non-mutually exclusive, there is some overlap between these events. The probability that A or B will occur is the sum of the probability of each event, minus the probability of the overlap.

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Multiplication Rule

Independent Events

Definition

Definition: Two events, A and B, are **independent** if the fact that A occurs does not affect the probability of B occurring.

Some other examples of independent events are:

- Landing on heads after tossing a coin **AND** rolling a 5 on a single 6-sided die.
- Choosing a marble from a jar **AND** landing on heads after tossing a coin.
- Choosing a 3 from a deck of cards, replacing it, **AND** then choosing an ace as the second card.
- Rolling a 4 on a single 6-sided die, **AND** then rolling a 1 on a second roll of the die.

Definition

- To find the probability of two independent events that occur in sequence, find the probability of each event occurring separately, and then multiply the probabilities.
- Note that multiplication is represented by AND.
- **Multiplication Rule 1:** When two events, A and B, are independent, the probability of both occurring is:

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

Example

- Experiment 1: A dresser drawer contains one pair of socks with each of the following colors: blue, brown, red, white and black. Each pair is folded together in a matching set. You reach into the sock drawer and choose a pair of socks without looking. You replace this pair and then choose another pair of socks.
- What is the probability that you will choose the red pair of socks both times?

Probabilities:

$$P(\text{red}) = \frac{1}{5}$$

$$\begin{aligned} P(\text{red and red}) &= P(\text{red}) \cdot P(\text{red}) \\ &= \frac{1}{5} \cdot \frac{1}{5} \\ &= \frac{1}{25} \end{aligned}$$

Example

Experiment 2: A coin is tossed and a single 6-sided die is rolled.

- Find the probability of landing on the head side of the coin and rolling a 3 on the die.

Probabilities:

$$
\begin{aligned}
P(\text{head}) &= \frac{1}{2} \\
P(3) &= \frac{1}{6} \\
P(\text{head and } 3) &= P(\text{head}) \cdot P(3) \\
&= \frac{1}{2} \cdot \frac{1}{6} \\
&= \frac{1}{12}
\end{aligned}
$$

Example

- Experiment 3: A card is chosen at random from a deck of 52 cards. It is then replaced and a second card is chosen.
- What is the probability of choosing a jack and then an eight?

Probabilities:

$$P(\text{jack}) = \frac{4}{52}$$

$$P(8) = \frac{4}{52}$$

$$\begin{aligned} P(\text{jack and } 8) &= P(\text{jack}) \cdot P(8) \\ &= \frac{4}{52} \cdot \frac{4}{52} \\ &= \frac{16}{2704} \\ &= \frac{1}{169} \end{aligned}$$

Example

- Experiment 4: A jar contains 3 red, 5 green, 2 blue and 6 yellow marbles. A marble is chosen at random from the jar. After replacing it, a second marble is chosen.
- What is the probability of choosing a green and then a yellow marble?

Probabilities:

$$P(\text{green}) = \frac{5}{16}$$

$$P(\text{yellow}) = \frac{6}{16}$$

$$P(\text{green and yellow}) = P(\text{green}) \cdot P(\text{yellow}) = \frac{5}{16} \cdot \frac{6}{16} = \frac{30}{256} = \frac{15}{128}$$

Note to Remember

- Each of the experiments above involved two independent events that occurred in sequence.
- In some cases, there was *replacement of the first item before choosing the second item; this replacement was needed in order to make the two events independent.*
- Multiplication Rule 1 can be extended to work for three or more independent events that occur in sequence.
- This is demonstrated in Experiment 5.

Example

Experiment 5: A school survey found that 9 out of 10 students like pizza. If three students are chosen at random with replacement, what is the probability that all three students like pizza?

Probabilities:

P(student 1 likes pizza) $= \frac{9}{10}$

P(student 2 likes pizza) $= \frac{9}{10}$

P(student 3 likes pizza) $= \frac{9}{10}$

P(student 1 and student 2 and student 3 like pizza) $= \frac{9}{10} \cdot \frac{9}{10} \cdot \frac{9}{10} = \frac{729}{1000}$

Dependent Events

Dependent Events

- **Definition:** Two events are **dependent** if the outcome or occurrence of the first affects the outcome or occurrence of the second so that the probability is changed.

Example

- Experiment 1: A card is chosen at random from a standard deck of 52 playing cards. Without replacing it, a second card is chosen.
- What is the probability that the first card chosen is a queen and the second card chosen is a jack?

Example

Probabilities:

P(queen on first pick) $= \frac{4}{52}$

P(jack on 2nd pick given queen on 1st pick) $= \frac{4}{51}$

P(queen and jack) $= \frac{4}{52} \cdot \frac{4}{51} = \frac{16}{2652} = \frac{4}{663}$

- Experiment 1 involved two compound, dependent events. The probability of choosing a jack on the second pick given that a queen was chosen on the first pick is called a *conditional probability*.

Conditional Probability

- The **conditional probability** of an event B in relationship to an event A is the probability that event B occurs given that event A has already occurred.
- The notation for conditional probability is P(B|A) [pronounced as *The probability of event B given A*].
- The notation used above does *not* mean that B is divided by A. It means *the probability of event B given that event A has already occurred*.
- To find the probability of the two dependent events, we use a modified version of *Multiplication Rule 1.*

Multiplication Rule 2

- **Multiplication Rule 2:** When two events, A and B, are dependent, the probability of both occurring is:

$$P(A \text{ and } B) = P(A) \cdot P(B|A)$$

Example

- Experiment 2: Mr. Parietti needs two students to help him with a science demonstration for his class of 18 girls and 12 boys. He randomly chooses one student who comes to the front of the room. He then chooses a second student from those still seated.
- What is the probability that both students chosen are girls?

Probabilities P(Girl 1 and Girl 2) = P(Girl 1) and P(Girl 2|Girl 1)

$$= \frac{18}{30} \cdot \frac{17}{29}$$

$$= \frac{306}{870}$$

$$= \frac{51}{145}$$

Example

- Experiment 3: In a shipment of 20 computers, 3 are defective. Three computers are randomly selected and tested.
- What is the probability that all three are defective if the first and second ones are not replaced after being tested?

Probabilities: P(3 defectives) =

$$\frac{3}{20} \cdot \frac{2}{19} \cdot \frac{1}{18} = \frac{6}{6840} = \frac{1}{1140}$$

Example

- Four cards are chosen at random from a deck of 52 cards without replacement. What is the probability of choosing a ten, a nine, an eight and a seven in order?

Probabilities: P(10 and 9 and 8 and 7) =

$$\frac{4}{52} \cdot \frac{4}{51} \cdot \frac{4}{50} \cdot \frac{4}{49} = \frac{256}{6{,}497{,}400} = \frac{32}{812{,}175}$$

Example

- Experiment 5: Three cards are chosen at random from a deck of 52 cards without replacement. What is the probability of choosing 3 aces?

Probabilities: P(3 aces) =

$$\frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50} = \frac{24}{132{,}600} = \frac{1}{5{,}525}$$

Summary:

Two events are dependent if the outcome or occurrence of the first affects the outcome or occurrence of the second so that the probability is changed.

The *conditional probability* of an event B in relationship to an event A is the probability that event B occurs given that event A has already occurred.

The notation for conditional probability is $P(B|A)$.

When two events, A and B, are dependent, the probability of both occurring is: *$P(A \text{ and } B) = P(A) \cdot P(B|A)$*

Solve

- At a community swimming pool, there are 2 managers, 8 lifeguards, 3 concession stand clerks and 2 maintenance people. If a person is selected at random, find the probability that the person is either a lifeguard or a manager.